

ALEXANDER OLAGUEZ

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EDUCATION

University of Illinois Urbana-Champaign - The Grainger College of Engineering

Expected: May 2028

Bachelor of Science in Computer Engineering

City Colleges of Chicago - Wilbur Wright College, School of Engineering

Graduated: May 2026

Associate in Engineering Science, Engineering Pathway

GPA: 3.48/4.0

Related Coursework

Object-Oriented Programming in C++ I & II

Circuit Analysis

Differential Equations

Data Structures & Algorithms

Linear Algebra & Discrete Mathematics

Engineering Physics I, II & III

TECHNICAL SKILLS

Programming: C++, Python, C, Linux/Bash

Tools & Platforms: High-Performance Computing (HPC), GPU Computing, Slurm, Jupyter, Perlmutter (NERSC), VS Code, LTspice, CAD / 3D Modeling, HTML

EXPERIENCE

Argonne Leadership Computing Facility (ALCF), Argonne National Laboratory

Aug 2026

Intro to High-Performance Computing Bootcamp Participant

St. Charles, IL

- Profiled larnd-sim, the open-source liquid-argon simulator used by the Deep Underground Neutrino Experiment (DUNE), on Perlmutter at NERSC using NVIDIA Nsight Systems and Nsight Compute.
- Ran the workflow in Python and Jupyter, submitted and monitored GPU jobs with Slurm, and created data visualizations of the simulation output.
- Team identified `get_adc_values()` as parallelizing inefficiently above four threads per block; presented the profiling section of the final technical presentation.

Illinois Semiconductor Workforce Network (ISWN) at UIC

May 2025 – Jun 2025

Semiconductor Research Intern

Chicago, IL

- Performed semiconductor wafer fabrication in a cleanroom under laboratory-safety and contamination-control protocols: photoresist application, photolithography, hard bakes, chemical processing, etching, and metrology.
- Designed a MEMS thermal actuator to a 4 μm minimum feature, with 85 $\times$ 85 μm anchors and a $\sim$ 400 $\times$ 400 μm active region.

PROJECTS

Two-Way Calendar Sync Tool

Jul 2026 – Sep 2026

Personal Project - Python, Google Calendar API, Microsoft Graph API, OAuth 2.0

- Built a Python tool that syncs events in both directions between Google Calendar and Outlook, tracking 214 events across 3 calendar pairs.
- Detects edits by hashing each event's fields against a saved state file, and resolves events changed on both sides by keeping the most recently modified version.
- Added a safeguard that skips a calendar pair when more than half its tracked events are missing from a fetch, so an API failure cannot propagate as mass deletions.

Self-Charging Kinetic Energy Harvesting Prosthetic

Sep 2023 – Dec 2023

3D Model Designer - CAD

- Designed the complete 3D CAD model for a six-student self-charging prosthetic-limb concept that harvests kinetic energy from the wearer's steps.
- Presented the design and engineering rationale to judges from academic and industry backgrounds.

LEADERSHIP & INVOLVEMENT

Institute of Electrical and Electronics Engineers @ Wright

Aug 2025 – May 2026

- Founding Officer - Co-founded and helped ratify Wright College's first IEEE student chapter, expanding electrical and computer engineering opportunities for 50+ students in its first year.
 - o Organized and led 8+ hands-on workshops covering circuit fundamentals, multimeter use, and prototyping.

HONORS & AWARDS

Illinois Space Grant Consortium Scholarship; CME Group Star Partnership Mayoral Award; Schreiber Scholarship; Options for the Future Scholarship